

Unit Test Documentation

Bug Severity Classification System

Hemasree

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1 Introduction

This document describes the unit tests implemented for the Bug Severity Classification System using `pytest`. The goal is to validate that all API endpoints function correctly under normal and failure scenarios.

2 Test Scope

The following components are tested:

- Authentication (Signup, Login)
- Prediction API
- Feedback API
- History Endpoint
- Statistics Endpoint
- Drift Detection Endpoint
- Admin APIs
- Prometheus Metrics Endpoint

3 Test Cases

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Test Case ID	Input	Expected Output
UT01 Signup user	Valid username, email, password	HTTP 200 or 400 if duplicate
UT02 Login success	Correct username and password	HTTP 200, token returned
UT03 Login failure	Invalid credentials	HTTP 400 error

header ID	Test Case	Input	Expected Output
UT04	Predict success	Valid summary + token	HTTP 200, severity + confidence returned
UT05	Predict without token	Missing Authorization header	HTTP 401 Unauthorized
UT06	Feedback submission	Valid prediction ID	HTTP 200, status = ok
UT07	Get history	Valid token	HTTP 200, list of predictions
UT08	Feedback stats	No input	HTTP 200, total + accuracy_pct fields
UT09	Drift check	No input	HTTP 200, drift_detected field
UT10	Admin stats	No input	HTTP 200, system statistics returned
UT11	Admin users	No input	HTTP 200, list of users
UT12	Metrics endpoint	No input	HTTP 200, Prometheus metrics text

4 Tools Used

- pytest
- FastAPI TestClient
- Python 3.12

5 Conclusion

All major API endpoints were tested successfully. Both valid and invalid scenarios were covered to ensure robustness of the system. These unit tests provide confidence that the application behaves as expected under typical usage conditions.